

## PROJECT1- Amazon sales

```
[3]: #Importing the required libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
[4]: #Loading the csv file
df=pd.read_csv("amazon_sales_data.csv")
```

```
[5]: df
```

|     | Order ID | Date             | Product          | Category      | Price       | Quantity | Total Sales | Customer Name | Customer Location | Payment Method | Status      |           |
|-----|----------|------------------|------------------|---------------|-------------|----------|-------------|---------------|-------------------|----------------|-------------|-----------|
|     | 0        | ORD0001          | 14 March 2025    | Running Shoes | Footwear    | 60       | 3           | 180           | Emma Clark        | New York       | Debit Card  | Cancelled |
|     | 1        | ORD0002          | 20 March 2025    | Headphones    | Electronics | 100      | 4           | 400           | Emily Johnson     | San Francisco  | Debit Card  | Pending   |
|     | 2        | ORD0003          | 15 February 2025 | Running Shoes | Footwear    | 60       | 2           | 120           | John Doe          | Denver         | Amazon Pay  | Cancelled |
|     | 3        | ORD0004          | 19 February 2025 | Running Shoes | Footwear    | 60       | 3           | 180           | Olivia Wilson     | Dallas         | Credit Card | Pending   |
|     | 4        | ORD0005          | 10 March 2025    | Smartwatch    | Electronics | 150      | 3           | 450           | Emma Clark        | New York       | Debit Card  | Pending   |
| ... | ...      | ...              | ...              | ...           | ...         | ...      | ...         | ...           | ...               | ...            | ...         |           |
| 245 | ORD0246  | 17 March 2025    | T-Shirt          | Clothing      | 20          | 2        | 40          | Daniel Harris | Miami             | Debit Card     | Cancelled   |           |
| 246 | ORD0247  | 30 March 2025    | Jeans            | Clothing      | 40          | 1        | 40          | Sophia Miller | Dallas            | Debit Card     | Cancelled   |           |
| 247 | ORD0248  | 05 March 2025    | T-Shirt          | Clothing      | 20          | 2        | 40          | Chris White   | Denver            | Debit Card     | Cancelled   |           |
| 248 | ORD0249  | 08 March 2025    | Smartwatch       | Electronics   | 150         | 3        | 450         | Emily Johnson | New York          | Debit Card     | Cancelled   |           |
| 249 | ORD0250  | 19 February 2025 | Smartphone       | Electronics   | 500         | 4        | 2000        | Emily Johnson | Seattle           | Amazon Pay     | Completed   |           |

250 rows × 11 columns

```
[6]: df.head(3)
```

[6]:

| Order ID | Date    | Product          | Category      | Price       | Quantity | Total Sales | Customer Name | Customer Location | Payment Method | Status     |           |
|----------|---------|------------------|---------------|-------------|----------|-------------|---------------|-------------------|----------------|------------|-----------|
| 0        | ORD0001 | 14 March 2025    | Running Shoes | Footwear    | 60       | 3           | 180           | Emma Clark        | New York       | Debit Card | Cancelled |
| 1        | ORD0002 | 20 March 2025    | Headphones    | Electronics | 100      | 4           | 400           | Emily Johnson     | San Francisco  | Debit Card | Pending   |
| 2        | ORD0003 | 15 February 2025 | Running Shoes | Footwear    | 60       | 2           | 120           | John Doe          | Denver         | Amazon Pay | Cancelled |

```
[7]: df.tail(3)
```

| [7]: | Order ID | Date             | Product    | Category    | Price | Quantity | Total Sales | Customer Name | Customer Location | Payment Method | Status    |
|------|----------|------------------|------------|-------------|-------|----------|-------------|---------------|-------------------|----------------|-----------|
| 247  | ORD0248  | 05 March 2025    | T-Shirt    | Clothing    | 20    | 2        | 40          | Chris White   | Denver            | Debit Card     | Cancelled |
| 248  | ORD0249  | 08 March 2025    | Smartwatch | Electronics | 150   | 3        | 450         | Emily Johnson | New York          | Debit Card     | Cancelled |
| 249  | ORD0250  | 19 February 2025 | Smartphone | Electronics | 500   | 4        | 2000        | Emily Johnson | Seattle           | Amazon Pay     | Completed |

Checking the Null values

```
[8]: df.shape
```

```
[8]: (250, 11)
```

```
[9]: df.isnull()
```

[9]:

|     | Order ID | Date  | Product | Category | Price | Quantity | Total Sales | Customer Name | Customer Location | Payment Method | Status |
|-----|----------|-------|---------|----------|-------|----------|-------------|---------------|-------------------|----------------|--------|
| 0   | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 1   | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 2   | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 3   | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 4   | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| ... | ...      | ...   | ...     | ...      | ...   | ...      | ...         | ...           | ...               | ...            | ...    |
| 245 | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 246 | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 247 | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 248 | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |
| 249 | False    | False | False   | False    | False | False    | False       | False         | False             | False          | False  |

250 rows × 11 columns

```
[10]: df.isnull().sum()
```

```
[10]: Order ID          0
      Date            0
      Product         0
      Category        0
      Price           0
      Quantity        0
      Total Sales     0
      Customer Name    0
      Customer Location 0
      Payment Method   0
      Status          0
      dtype: int64
```

```
[16]: df.describe()
```

[16]:

|       | Price       | Quantity   | Total Sales |
|-------|-------------|------------|-------------|
| count | 250.000000  | 250.000000 | 250.000000  |
| mean  | 343.580000  | 2.856000   | 975.380000  |
| std   | 380.635808  | 1.429489   | 1252.112254 |
| min   | 15.000000   | 1.000000   | 15.000000   |
| 25%   | 40.000000   | 2.000000   | 100.000000  |
| 50%   | 150.000000  | 3.000000   | 400.000000  |
| 75%   | 600.000000  | 4.000000   | 1500.000000 |
| max   | 1200.000000 | 5.000000   | 6000.000000 |

```
[17]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250 entries, 0 to 249
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Order ID              250 non-null   object
1   Date                  250 non-null   object
2   Product               250 non-null   object
3   Category              250 non-null   object
4   Price                 250 non-null   int64
5   Quantity              250 non-null   int64
6   Total Sales           250 non-null   int64
7   Customer Name         250 non-null   object
8   Customer Location     250 non-null   object
9   Payment Method        250 non-null   object
10  Status                250 non-null   object
dtypes: int64(3), object(8)
memory usage: 21.6+ KB

```

## Plotting

```

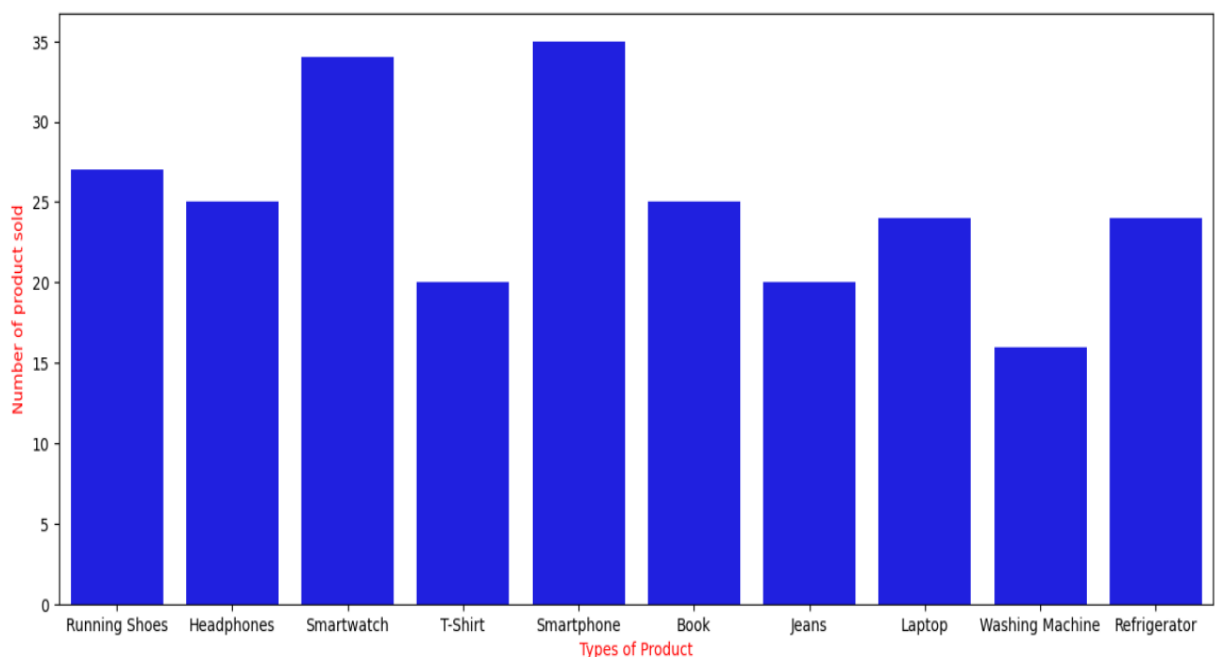
[65]: plt.figure(figsize=(15,6))
      sns.countplot(x=df["Product"],color="blue")
      plt.xlabel("Types of Product",color="red")
      plt.ylabel("Number of product sold",color="red")

```

```

[65]: Text(0, 0.5, 'Number of product sold')

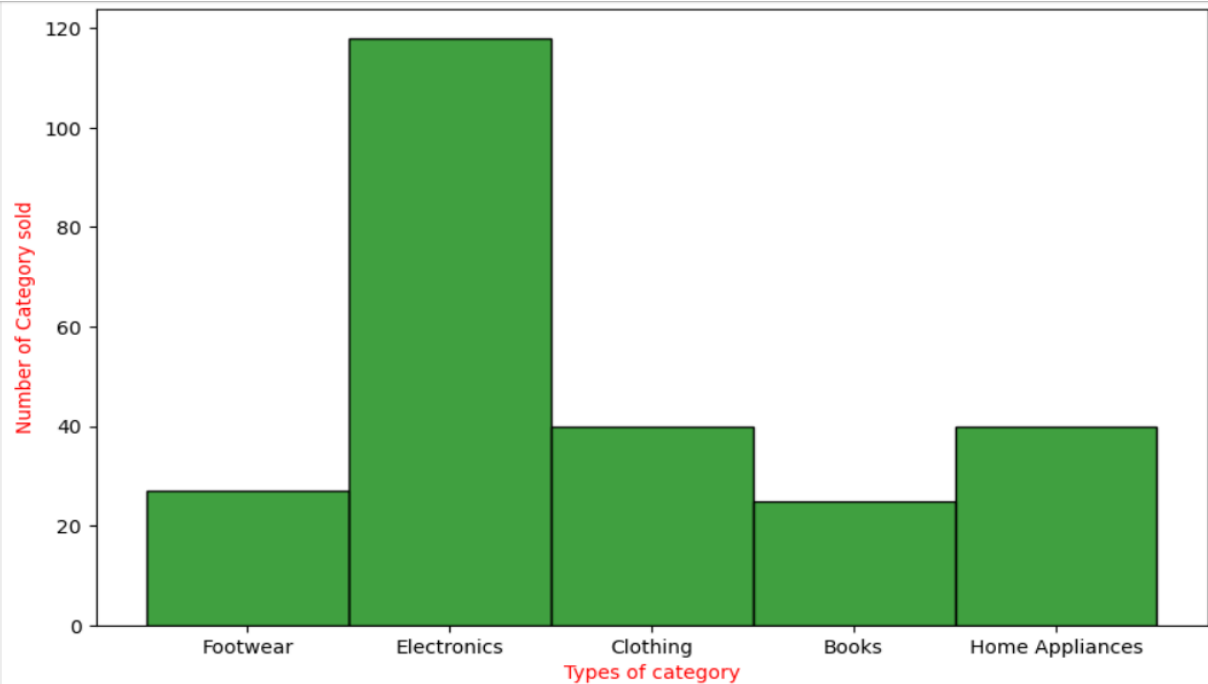
```



Insight- "Our top-performing products, smartphones and smartwatches are driving the majority of sales."

```
[66]: plt.figure(figsize=(10,6))
sns.histplot(x=df["Category"],color="green")
plt.xlabel("Types of category",color="red")
plt.ylabel("Number of Category sold",color="red")
```

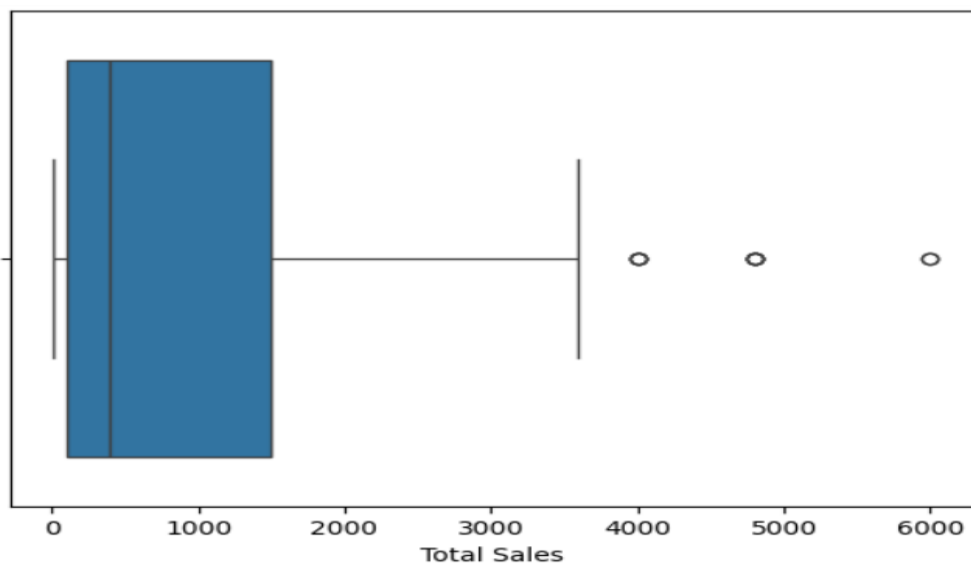
```
[66]: Text(0, 0.5, 'Number of Category sold')
```



Insight- "Electronics are in high demand, driving strong sales."

```
[67]: sns.boxplot(x="Total Sales", data=df)
```

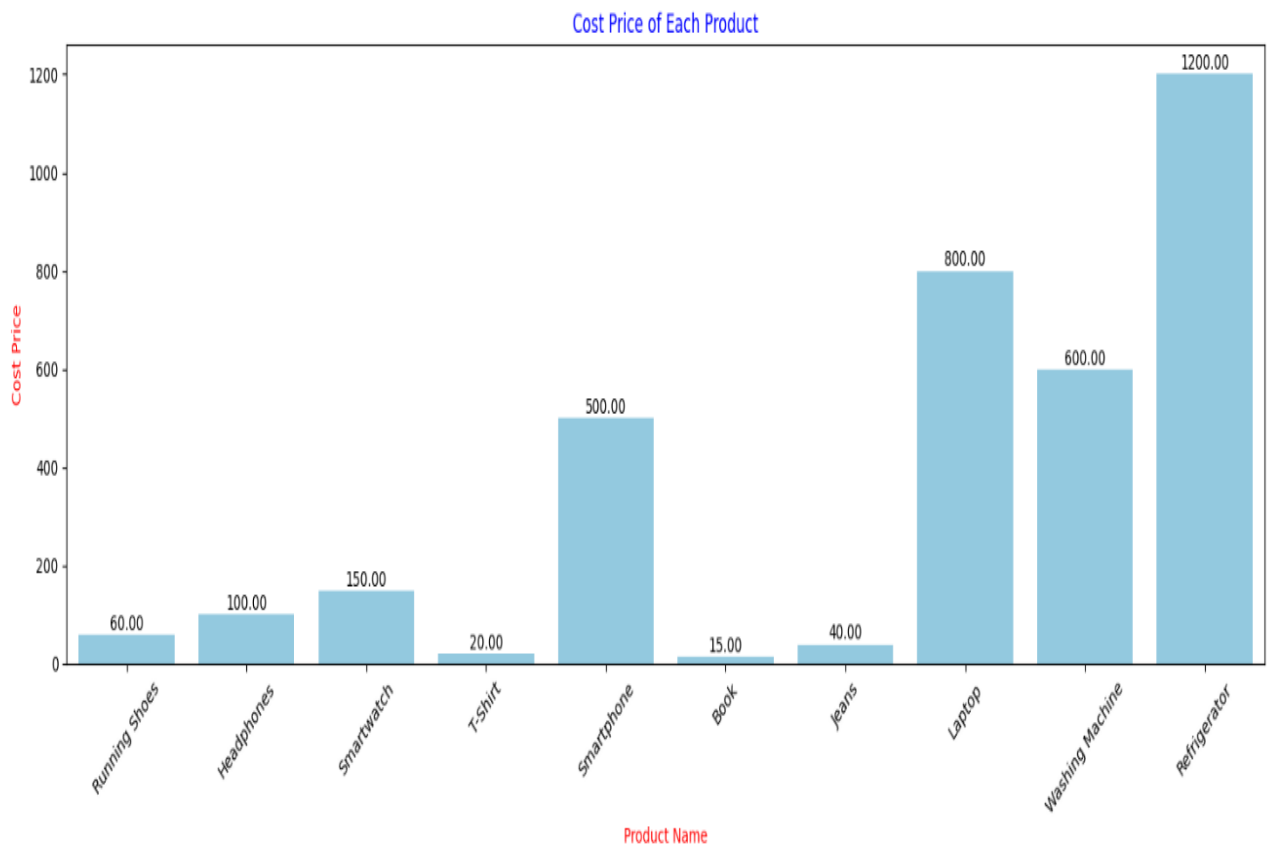
```
[67]: <Axes: xlabel='Total Sales'>
```



```
[68]: plt.figure(figsize=(15, 6))
ax = sns.barplot(x="Product", y="Price", data=df, color="skyblue")

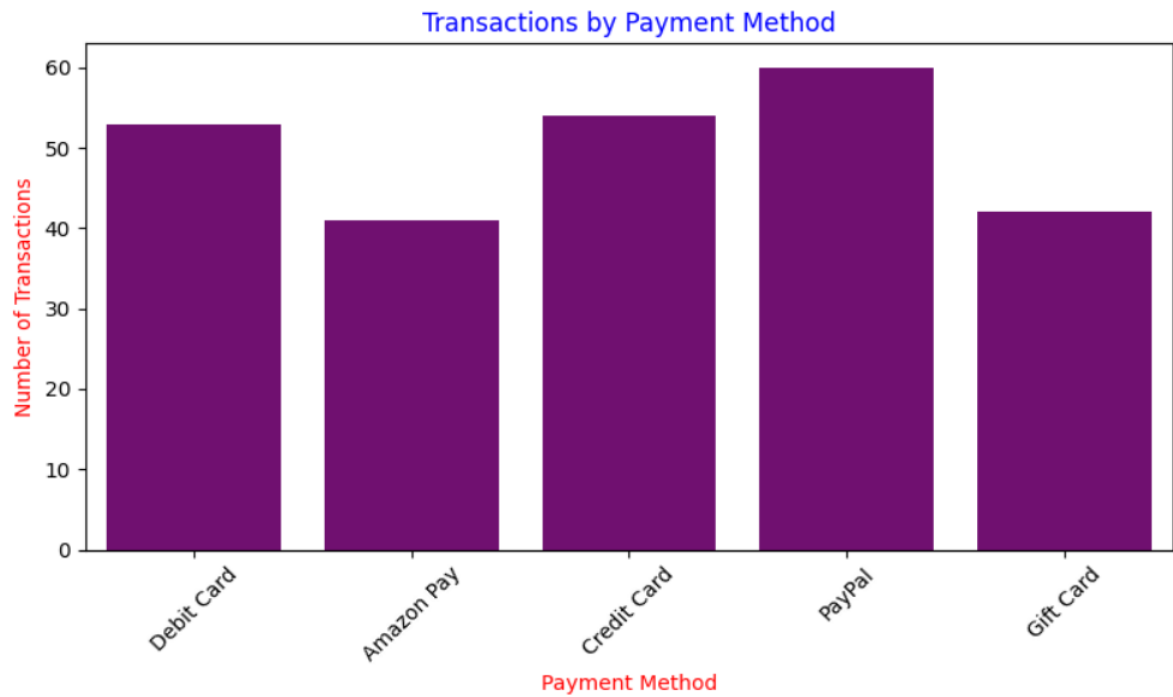
# Add labels on top of each bar
for p in ax.patches:
    ax.text(p.get_x() + p.get_width() / 2,
            p.get_height() + 0.5, # adjust vertical position
            f'{p.get_height():.2f}', # format the price
            ha='center', va='bottom', color='black', fontsize=10)

plt.xlabel("Product Name", color="red")
plt.ylabel("Cost Price", color="red")
plt.title("Cost Price of Each Product", color="blue")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



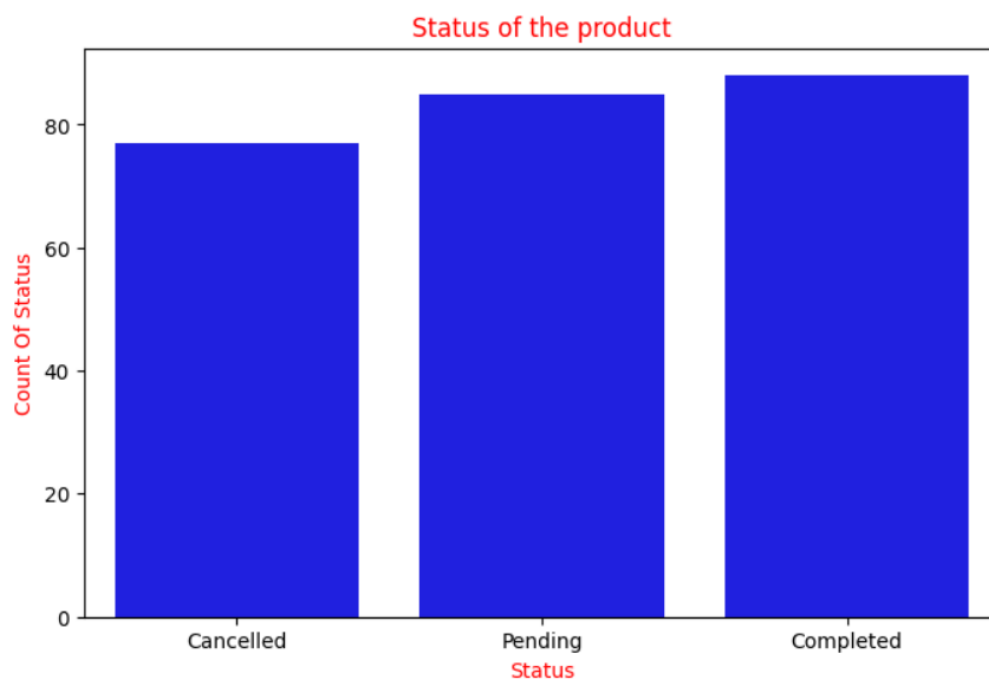
```
[71]: plt.figure(figsize=(8, 5))
sns.countplot(x="Payment Method", data=df, color="purple")

plt.xlabel("Payment Method", color="red")
plt.ylabel("Number of Transactions", color="red")
plt.title("Transactions by Payment Method", color="blue")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



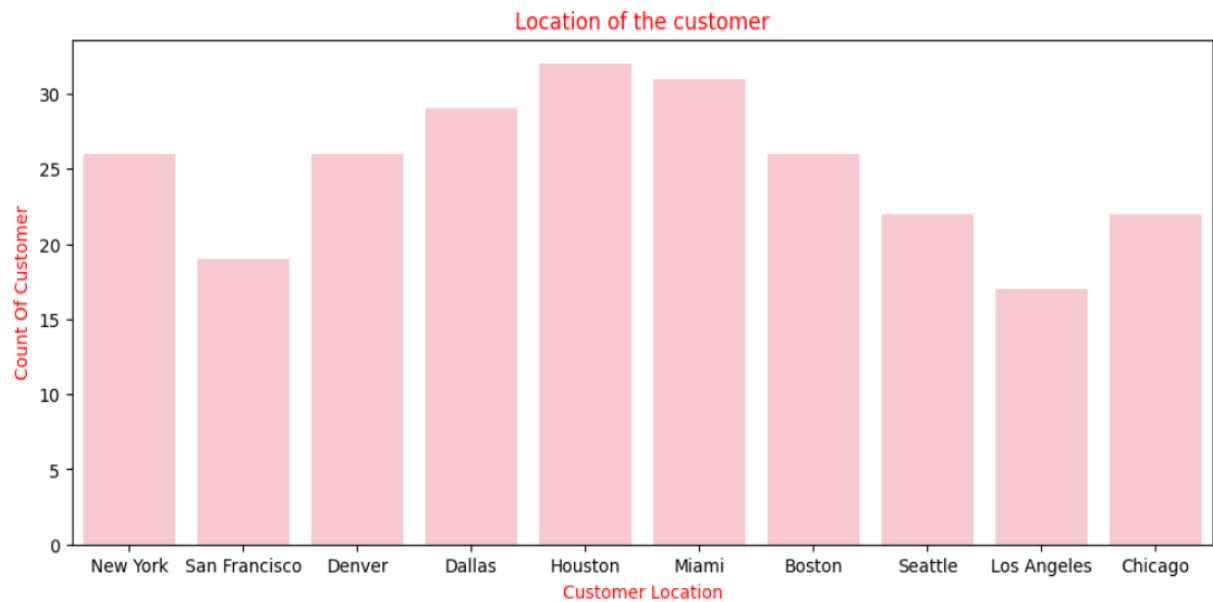
Insight- "The majority of customers prefer 'PayPal' as their payment method, followed by 'credit cards', 'debit cards', and 'Amazon Pay', 'Gift cards', however, remain the least preferred option.

```
[82]: plt.figure(figsize=(8, 5))
sns.countplot(x="Status", data=df, color="blue")
plt.xlabel("Status", color="red")
plt.ylabel("Count Of Status", color="red")
plt.title("Status of the product", color="red")
plt.show()
```



Insight- "The majority of product statuses are marked as 'Completed,' indicating that most transactions are successfully processed and fulfilled".

```
[83]: plt.figure(figsize=(12, 5))
sns.countplot(x="Customer Location", data=df, color="pink")
plt.xlabel("Customer Location", color="red")
plt.ylabel("Count Of Customer", color="red")
plt.title("Location of the customer", color="red")
plt.show()
```



Insight- "The majority of top customers are from Houston, Miami, and Mallad, indicating a strong customer base in these regions."